L23: Comparing two population means

Recap

- Last lecture we introduced the t test, which relaxes some of the assumptions we needed for the z test.
- Both the z test and the t test we have learned so far have compared a sample to a hypothesized mean.
- These are one sample tests, meaning we have one variable of interest, and we take one sample. – Using this one sample, we're interested in inferring the (unknown) mean of a population from which the sample was drawn – e.g. whether the population mean differs from some null value ($H_0 : \mu = 68mm$)

Learning objective for today

Today we will look at two-sample t-test

- Comparing means from two populations

Comparing means from two populations

Comparing two population means

- In many types of public health studies, rather than comparing one sample to a hypothesized value, we are interested in comparing two samples from two different populations.
- We want to shed light on the question: Do these two samples come from two groups with different underlying means? – e.g., $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_a : \mu_1 - \mu_2 \neq 0$ (two-sided alternative)

Comparing two population means

We like to compare two means! In public health, we like to:

- run randomized controlled trials where we compare a treated group to a placebo group. Do their mean health outcomes differ?
- conduct observational studies where we have exposed and unexposed individuals. Do their mean health outcomes differ?

This is covered in chapter 18 in your book. Note that we are skipping around a bit.

Comparing two population means, graphically

- Make two histograms, one for each sample
- Compare their shapes, centers (means or medians) and spreads (standard deviations)
- Could instead make two box plots and compare their medians and IQRs

Example with heights from two different populations

Are the heights of US and Dutch-born men different?

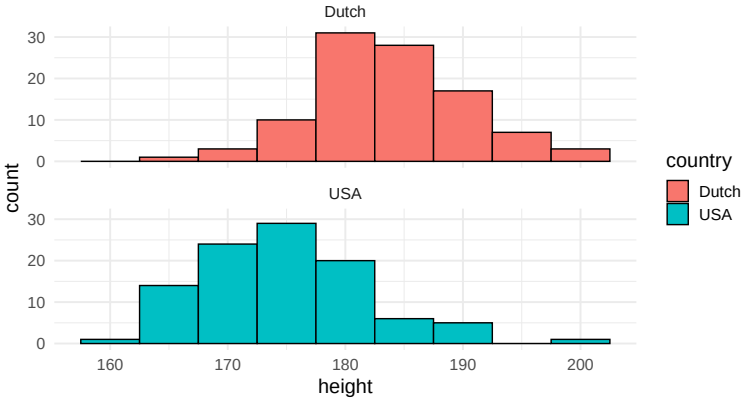
```
## # A tibble: 2 x 4
##   country sample_mean sample_sd length
##   <chr>         <dbl>     <dbl>  <int>
## 1 Dutch          184.      6.75    100
## 2 USA            175.      7.16    100
```

Example with heights from two different populations

Plotting two distributions:

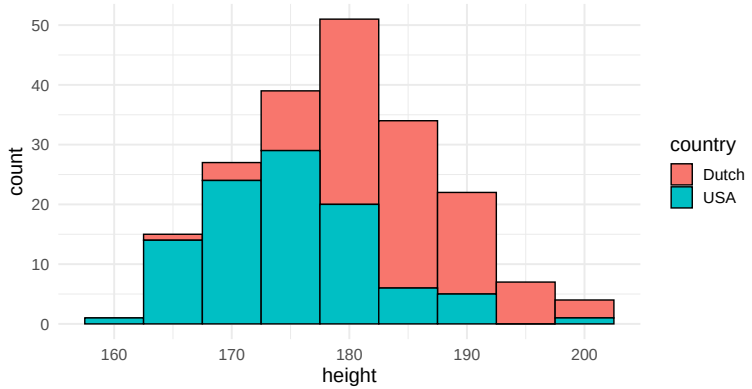
```
ggplot(height_data, aes(x = height)) +
  geom_histogram(aes(fill = country), binwidth = 5, col = "black") +
  theme_minimal(base_size = 15) +
  facet_wrap(~country, nrow = 2)
```

Example with heights from two different populations



Example with heights from two different populations

If we take out the last line of that code, the histograms will be plotted on one grid. Notice what happens to the columns...

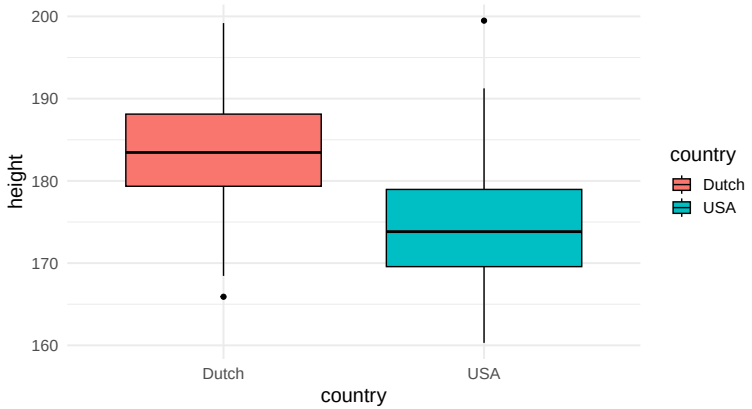


Example with heights from two different populations

Revisiting the code for box plots:

```
ggplot(height_data, aes(x = country, y = height)) +
  geom_boxplot(aes(fill = country), col = "black") +
  theme_minimal(base_size = 15)
```

Example with heights from two different populations



Two sample t test

Independent or non-independent samples?

An important requirement for using a basic t-test as a tool for this comparison, is that the observations from the two groups are independent from one another.

- If there is a relationship between the two groups we will use a different type of t-test, the paired t-test, which we will introduce later.

When we compare these two samples, we also assume that both populations were sampled in the same (random) way and that the measurement of the variable we are comparing was done in the same way.

Conditions for inference comparing two means

- Both populations are Normally distributed. The means and standard deviations of the populations are unknown.
- In practice, it is enough that the distributions have similar shapes and that the data have no strong outliers.

Notation

Notation for the population parameters

Population	Variable	Population mean	Population SD
1	x_1	μ_1	σ_1
2	x_2	μ_2	σ_2

Notation for the sample statistics

Population	Sample size	Sample mean	Sample SD
1	n_1	$\bar{x_1}$	s_1
2	n_2	$\bar{x_2}$	s_2

To perform inference about the difference $\mu_1 - \mu_2$ between the means of the two populations, we start from the difference $\bar{x}_1 - \bar{x}_2$ between the means of the two samples.

Two-sample t procedures

- With one-sample procedures, we had one $\bar{x}$ and we would draw the sampling distribution for $\bar{x}$. It was centered at μ with standard error $\sigma/\sqrt{n}$
- With two samples, we have two sample averages $\bar{x}_1$ and $\bar{x}_2$.
- What do we do?

Example with heights from two different populations

Are the heights of US and Dutch-born men different?

$\bar{x}_{USA} = 174.7042$ and $\bar{x}_D = 183.6690$ so the difference is $\bar{x}_D - \bar{x}_{USA} = 8.9648$

What would happen if we took another set of two samples of USA and Dutch-born men?

- We would expect these sample means to change.
- We could draw an approximate sampling distribution for the difference between these two means.

The sampling distribution of $\bar{x}_1 - \bar{x}_2$

The distribution of the difference between two independent random variables has

- a mean equal to the difference of their respective means, and
- a variance equal to the sum of their respective variances.

That is:

- The mean of the sampling distribution for $\bar{x}_1 - \bar{x}_2$ is $\mu_1 - \mu_2$.
- The standard deviation of the sampling distribution is: $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Our *estimate* of the standard deviation of the sampling distribution is:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Recall the one sample t-test!

$$\frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

We need to generalize this by replacing each piece in the t-test by the calculations on the previous slide.

For the null hypothesis $H_0 : \mu_1 - \mu_2 = \delta_0$, the two-sample t-test is therefore:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{SE} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under the null, the two-sample t statistic has approximately a t distribution. The approximation is accurate when both sample sizes are greater than or equal to 5.

Degrees of freedom for the two-sample t-test...

is bananas.

$$df = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{\frac{1}{n_1 - 1}(\frac{s_1^2}{n_1})^2 + \frac{1}{n_2 - 1}(\frac{s_2^2}{n_2})^2}$$

Often this is approximated by assuming that the degrees of freedom is $n_1 - 1$ or $n_2 - 1$, whichever is smaller, or by making an assumption that the variance in the two samples is equal and approximating the df with $n_1 + n_2 - 2$

We will NOT calculate df by hand - we will use R

Hypothesis testing when you have two samples

$H_0 : \mu_1 - \mu_2 = 0$, obtain the two-sample t-test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where the test p-value is the probability, when H_0 is true, of getting a test statistic t at least as extreme in the direction of H_a as that obtained, and is computed as the corresponding area under the t distribution with the appropriate degrees of freedom.

Example, continued

Let R do the work for you:

```
usa_ht <- height_data_wide %>% pull(usa)
dutch_ht <- height_data_wide %>% pull(dutch)

t.test(usa_ht, dutch_ht, alternative = "two.sided")
```

```
##
##  Welch Two Sample t-test
##
## data:  usa_ht and dutch_ht
## t = -9.1103, df = 197.35, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -10.905418  -7.024282
## sample estimates:
## mean of x mean of y
```

Example, continued

Note that `t.test` gives you both the t-test results (t-statistic (called "t" in the output), df, and p-value), as well as the 95% CI. We got both because we performed a two-sided test.

Example: Zika vaccine 2017 article

ORIGINAL ARTICLE

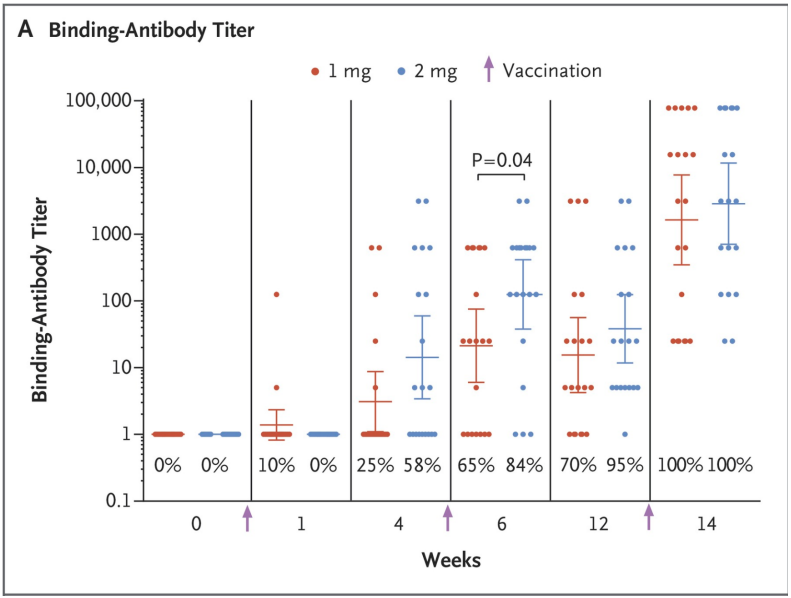
Safety and Immunogenicity of an Anti–Zika Virus DNA Vaccine — Preliminary Report

Pablo Tebas, M.D., Christine C. Roberts, Ph.D., Kar Muthumani, Ph.D., Emma L. Reuschel, Ph.D., Sagar B. Kudchodkar, Ph.D., Faraz I. Zaidi, M.S., Scott White, M.D., Amir S. Khan, Ph.D., Trina Racine, Ph.D., Hyeree Choi, B.S., Jean Boyer, Ph.D., Young K. Park, J.D., et al.

Example: Zika vaccine

STATISTICAL ANALYSIS The antibody-binding response that was assessed on ELISA is reported as the proportion of participants in whom an antibody response developed at a given time point and as the geometric mean titer (both with 95% confidence intervals). We used Fisher's exact test to determine positive response rates and Student's t-test to compare the magnitude of the log-transformed antibody response between the two dose groups and within individuals as the change from baseline.

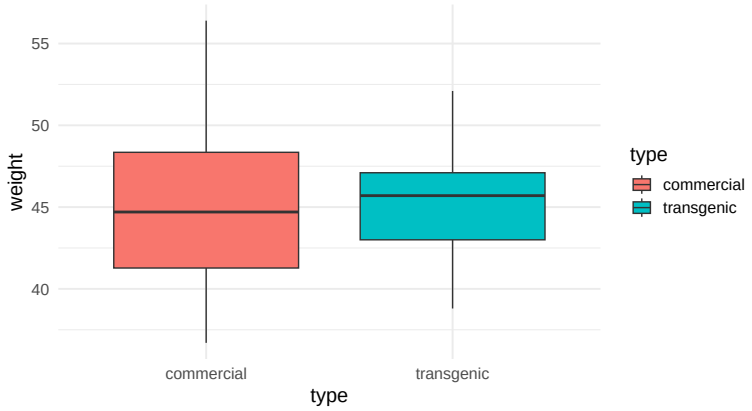
Example: Zika vaccine



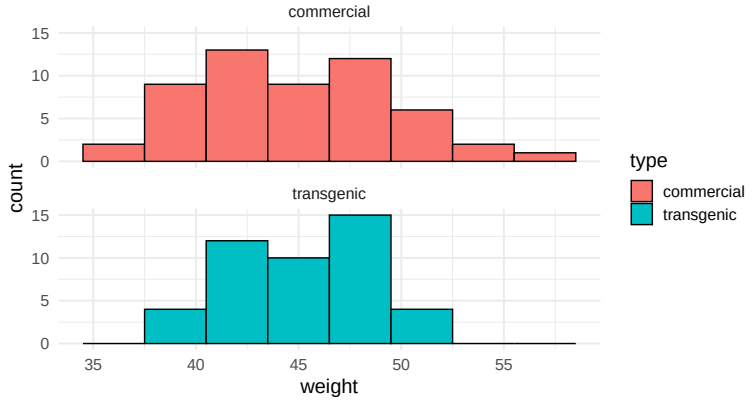
Example: Transgenic chickens

Infection of chickens with the avian flu is a threat to both poultry production and human health. A research team created transgenic chickens resistant to avian flu infection. Could the modification affect the chicken in other ways? The researchers compared the hatching weights (in grams) of 45 transgenic chickens and 54 independently selected commercial chickens of the same breed.

Example: Transgenic chickens



Example: Transgenic chickens



Estimate the size of the difference between the two means

```
means <- chicken_data %>%
  group_by(type) %>%
  summarise(mean_weight = mean(weight))

diff_means <- means[1, 2] - means[2, 2]
diff_means
```

```
##   mean_weight
## 1  -0.1533333
```

The estimated mean difference is -0.153 grams.

Estimate the standard error

```
chicken_stats <- chicken_data %>%
  group_by(type) %>%
  summarise(mean_weight = mean(weight),
            sd_weight = sd(weight),
            n = length(weight))
chicken_stats
```

```
## # A tibble: 2 x 4
##   type       mean_weight sd_weight     n
##   <chr>            <dbl>     <dbl> <int>
## 1 commercial        45.0      4.57    54
## 2 transgenic        45.1      3.32    45
```

Use the output to calculate the SE:

$$SE = \sqrt{\frac{4.568872^2}{54} + \frac{3.320836^2}{45}} = 0.7947528 \text{ (recall } SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\text{)}$$

Calculate the t-statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$t = \frac{(44.98889 - 45.14222) - (0)}{0.7947528} = -0.1929279$$

What is the chance of observing the t-statistic -0.193 on the t-distribution with the appropriate degrees of freedom?

To answer this, we would need to calculate the degrees of freedom using that crazy formula. We won't do this. Instead, we will ask R to do the test for us (and verify that our calculated t-statistic matches R's test)

t.test in R

Pay attention to the arguments specified by `t.test`. The first argument is the weight data for the commercial chickens and the second argument is the weight data for the transgenic chickens.

```
t.test(commercial_weight, transgenic_weight, alternative = "two.sided")
```

```
##
##  Welch Two Sample t-test
##
## data:  commercial_weight and transgenic_weight
## t = -0.19293, df = 95.344, p-value = 0.8474
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -1.731044  1.424377
## sample estimates:
## mean of x mean of y
##  44.98889  45.14222
```

Robustness of the two-sample t procedures

- These procedures are more robust than the one-sample t procedures, especially if the data are skewed.
- When the sizes of the two samples are equal and the two populations being compared have similar shapes, the t procedures will work well for sample sizes as small as $n_1 = n_2 = 5$.
- When the two populations have different shapes, larger samples are needed (e.g., one skewed left and the other skewed right).

CI for two sample t-test

Confidence intervals for the two-sample t-test

For a one sample t-test the CI looked like this:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

When we have two samples it will look like this:

$$(\bar{x_1} - \bar{x_2}) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t^* is the critical value with area C between $-t^*$ and t^* under the t density curve with the appropriate degrees of freedom.

Summary

A two sample t-test takes the form:

t.test(first sample data, second sample data, alternative = greater, less or two.sided)

which can be used to compare two population means and test whether their difference is equal to some null value.

Parting Humor (from XKCD.com)

